Week 10 Final:

AI-Assisted Triage Urgency Messages

The interface uses four urgency levels: Critical, High, Medium and Low. Each message uses plain language, an action verb and a clear indication of what the nurse should do next. Colour is used as a supporting visual cue rather than the sole indicator of urgency.

🔴 CRITICAL = ACT NOW: Assess this patient immediately.

This means that the patient requires immediate attention.

🟠 HIGH = ASSESS SOON: Prioritise this patient for assessment.

Communicates that the patient should be assessed promptly.

🟡 MEDIUM = REVIEW: Assess this patient when available.

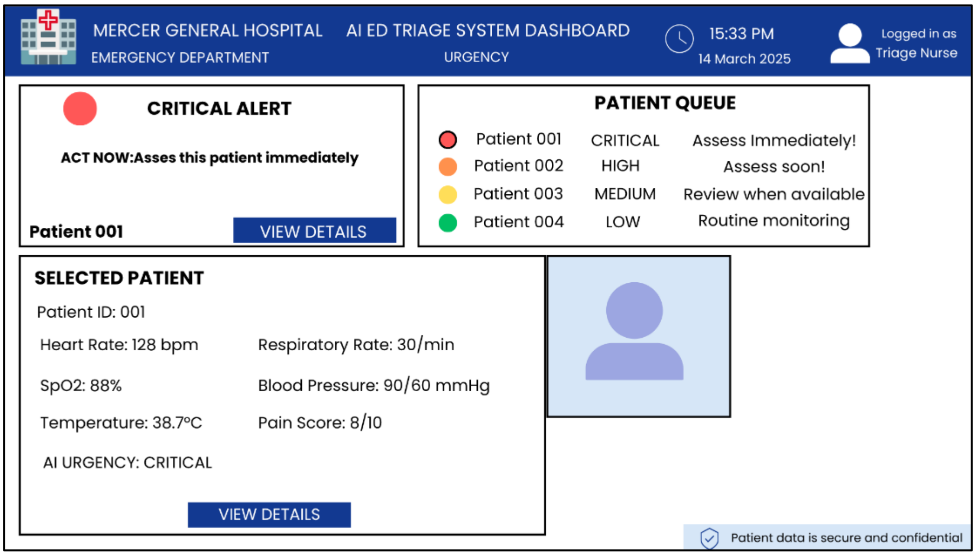
Communicates that the patient requires review but is not the highest priority.

🟢 LOW = MONITOR: Continue routine triage and monitoring.

Communicates that routine monitoring and triage should continue.

Initial HCI Prototype ED Triage Dashboard

Purpose:
The prototype illustrates how an AI-assisted triage system could communicate patient urgency to a triage nurse through a screen-based interface.



The prototype includes:

1. Patient queue - displays patients awaiting assessment.
2. Urgency indicator - identifies the urgency level of each patient.
3. Alert banner - displays the most important AI-generated recommendation.
4. View Details button - allows the nurse to access additional patient information.
5. Patient information panel - provides relevant physiological information supporting the alert.

Accessibility Considerations for the HCI Prototype

The interface is designed for use in a busy emergency-department environment where users may be working under time pressure and fatigue.

1. Colour blindness
    - Concern: Users may not be able to distinguish urgency levels based on colour alone.
    - Design response:
      Each alert contains a written urgency label, **CRITICAL, HIGH, MEDIUM or LOW**, as well as an explicit action message. Colour

therefore provides an additional cue rather than being the only source of information.

2. Cognitive load
    - Concern: A triage nurse may be working under pressure, particularly during a busy or late shift. Long or complicated messages could slow interpretation.

    - Design response: Alert messages are short, use plain language and begin with a clear action. The interface presents the most important information prominently rather than requiring the user to read large amounts of text.

3. Screen Readability
    - Concern: Small text, poor contrast or a cluttered interface could make important information difficult to identify quickly.

    - Design response: The prototype uses large headings, clear visual separation, readable text and a consistent layout. Urgency information is placed prominently so that it can be identified quickly.

Peer Testing Plan

Tester	Correct ranking?	Colour helpful?	Interface confusing?	Suggested improvement
Tester 1	Yes	Yes	No	No changes suggested
Tester 2	Yes	Yes	No	No changes suggested
Tester 3	Yes	Yes	No	Suggested changing the symbols

Results:
All three testers correctly ranked the four urgency messages from most urgent to least urgent. All three said that the colour coding helped them understand the urgency levels, and none found the interface confusing.

One tester suggested that the symbols/icons used in the interface could be changed, giving you one concrete improvement to consider for the next iteration.

Next Iteration:
Based on the peer-testing results, the next iteration should review the symbols used

alongside the urgency indicators. Different symbols can be considered and tested to determine whether they improve recognition without adding unnecessary visual complexity.

Further testing should also involve more participants. While cohort members provided useful initial feedback, they cannot completely reproduce the workload, interruptions, fatigue and clinical knowledge of an actual Emergency Department triage nurse. Future testing should therefore include healthcare professionals where appropriate.

Additional testing should measure how quickly users identify a Critical patient, whether they can correctly interpret the urgency levels while distracted, and whether they understand that the AI recommendation is only a decision-support recommendation.

Conclusion

The four urgency messages were designed to be clear, actionable and accessible under pressure. Peer testing showed that all three testers correctly understood the urgency levels, found the colour coding helpful, and experienced no interface confusion. Feedback on the symbols will guide the next design iteration. Overall, the results support further refinement and testing of the prototype before any real-world clinical deployment.